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MOLD ASSEMBLIES FOR USE IN CREATING REPLICAS OF A SURFACE OF A COMPONENT

Abstract

A mold for use in creating a surface replication of a surface of a component including a body including a first bracket and an opposite second bracket. The first bracket and the second bracket each include an inner surface that defines at least a portion of a boundary of a mold cavity. The mold includes an adjustment mechanism for selectively coupling the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component.

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Background/Summary

TECHNICAL FIELD

[0001] The field of the disclosure relates generally to mold assemblies, and more particularly to mold assemblies for use in creating surface replicas of a component, such as a turbine blade.

BACKGROUND

[0002] Surface replicas are commonly used to analyze surface features, e.g., defects, pitting, cracks, or other signs of wear, of a surface of a component, such as a blade of a turbine or compressor. Surface replicas may be formed by applying a layer of a solidifying compound, in liquid form, over the surface of the component and after the compound solidifies, the solidified replica may be inspected using optical microscopy, for example, or any other known inspection method. Conventionally, the solidifying compound is dispensed across the surface in a free form manner using an injection device. The liquid compound may have a viscosity that enables the compound to remain generally in the applied area on the surface of the component while it solidifies.

[0003] If the surface replica is created properly, i.e., the resulting solidified compound has a sufficient and/or uniform thickness, the replica may be able to preserve surface features. However, when the liquid solidifying compound is applied conventionally in a free form manner, particularly across edges of components, the resulting replica may not have a sufficient thickness, and/or if the liquid solidifying compound pools away from the edge, the resulting replica may crack and/or be unusable. As such, because of the potential of high quality losses, the use of conventional methods of creating surface replicas may be limited.

[0004] Accordingly, a need exists for a system and method of forming surface replicas that produces a resulting replica having a greater consistency and uniformity that facilitates preventing and/or reducing quality losses.

SUMMARY

[0005] In one aspect, a mold for use in creating a surface replication of a surface of a component is provided. The mold includes a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface. The first and second inner surfaces define at least a portion of a boundary of a mold cavity. The mold assembly includes an adjustment mechanism for selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component.

[0006] In another aspect, a mold system for use in creating a surface replication of a surface of a component is provided. The mold system including a reservoir for use in storing a solidifying compound, an injection tool fluidically connected to the reservoir, and a mold. The mold includes a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface. The first and second inner surfaces define at least a portion of a boundary of a mold cavity. The mold includes an adjustment mechanism for selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component.

[0007] In another aspect, a method of creating a surface replication of a surface of a component is provided. The method includes providing a mold assembly defining a mold cavity, the mold assembly including a first bracket, a second bracket, and an adjustment mechanism for selectively increasing or decreasing a distance defined between the first and second brackets and adjusting the adjustment mechanism to selectively increase a distance defined between the first bracket and the

second bracket of the mold assembly. The method includes inserting a portion of a component between the first bracket and the second bracket and adjusting the adjustment mechanism to cause the mold assembly to be in sealing contact with a portion of the surface of the component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0009] FIG. 1 is a front perspective view of an exemplary mold assembly.

[0010] FIG. 2 is a rear perspective view of the mold assembly shown in FIG. 1.

[0011] FIG. 3 is a side view of the mold assembly shown in FIG. 1.

[0012] FIG. 4 is a front view of the mold assembly shown in FIG. 1.

[0013] FIG. 5 is a rear view of the mold assembly shown in FIG. 1.

[0014] FIG. 6 is a top view of the mold assembly shown in FIG. 1.

[0015] FIG. 7 is a bottom view of the mold assembly shown in FIG. 1.

[0016] FIG. 8 is a cross-sectional view of the mold assembly shown in FIG. 1.

[0017] FIG. 9 is a perspective view of an exemplary mold assembly coupled with a surface of an example turbine blade.

[0018] FIG. 10 is a perspective view of exemplary locations that the mold assembly shown in FIG. 1 may be coupled with a surface of an example turbine blade.

[0019] FIG. 11 is an example surface replica created using the mold assembly shown in FIG. 1

[0020] FIG. 12 is a perspective view of an exemplary solidifying compound injection device that may be used with the mold assembly shown in FIG. 1.

[0021] FIG. 13 is a process flow diagram of an exemplary method that may be used to create a surface replica using the mold assembly shown in FIG. 1.

[0022] Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of the disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of the disclosure. As such, the drawings are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

DETAILED DESCRIPTION

[0023] In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur and that the description includes instances where the event occurs and instances where it does not. Furthermore, references to “one embodiment” are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments “including” or “having” an element or a plurality of elements having a particular property may include additional such elements not having that property.

[0024] In exemplary embodiments described herein, a mold assembly, and the systems and methods associated therewith, may be used to create a replica of a surface of a component, referred to herein as a surface replica. The mold assembly defines at least partially, a portion of a boundary of a mold cavity, such that when the mold assembly is coupled with the component, the surface of the component also defines a portion of the boundary of the mold cavity. Surface replicas may be formed from a solidifying compound that is injected into the mold cavity in liquid form and is

subsequently solidified to create a replica of the surface which may be inspected to detect surface defects and/or wear properties of the component, such as, for example and without limitation, pitting, erosion, cavitation, rubbing, impacts, nicks, and/or dents. The mold assembly is oriented such that the mold cavity forms a surface replica having a suitable thickness that is capable of capturing surface defects. The mold assembly may be used to form surface replicas having a generally uniform and/or consistent or symmetric thickness such that the surface replicas are less likely to break or crack. Replicas that are too thin may crack or break and/or may be too thin to effectively capture surface defects. The mold assembly, and the systems and methods associated therewith, produce surface replicas that have a suitable thickness that enables the surface replicas to capture surface details with improved resolution and accuracy.

[0025] Embodiments of the mold assembly are particularly useful for creating surface replicas for components that have well defined edges and/or borders. In embodiments described herein, the mold assembly is selectively adjustable to enable the mold assembly to be coupled over an edge and/or across component surfaces of different shapes and/or sizes. In some embodiments, the mold assembly may be transported to a component, e.g., to a turbine blade of a gas turbine engine, and the mold may be selectively coupled to and/or uncoupled from a turbine blade to enable a surface replica of the surface or edge of the turbine blade to be formed. In some embodiments, the mold is a clip-on mold that includes an adjustment mechanism that may be used to selectively couple and uncouple the mold to a component by hand without the use of additional tools. For example, in some embodiments, the mold may be selectively coupled to or uncoupled from the component without the use of any tools, such as wrenches or drills.

[0026] Referring now to the drawings, FIG. 1 is a perspective view of an exemplary mold assembly, indicated generally at **100**. In the exemplary embodiment, the mold assembly **100** includes a body **102** including a first bracket **104** and an opposing second bracket **106** spaced a distance d away from the first bracket **104** such that a body opening **108** is defined. The mold assembly **100** also includes an adjustment mechanism **110** for selectively adjusting the relative position of the first bracket **104** and the second bracket **106**, e.g., the adjustment mechanism **110** may be selectively used to widen or narrow the distance d defined between the first and second brackets **104** and **106**, respectively. The first bracket **104** and the second bracket **106** may define, at least partially, a boundary **112** of a mold cavity **114**, as described in more detail below.

[0027] In embodiments described herein, the adjustment mechanism **110** enables the mold assembly **100** to be selectively coupled to, and/or uncoupled from, a component **200**, as best seen in FIGS. 9 and 10, for example. Selectively coupling the adjustment mechanism **110** to a component **200** enables a surface replica **300** formed of a solidifying compound **302**, as best seen in FIG. 11, of a component surface **202** to be created. In some embodiments, the mold assembly **100** may be selectively couplable to any suitable component **200** to form a surface replica **300**, for example and without limitation, a compressor blade, a turbine blade, a vane of an inlet guide, and the like, wherein resulting surface replica **300** has a consistent and suitable thickness that facilitates reducing quality losses and that facilitates an accurate and detailed surface replica **300** being created. During formation of a surface replica **300**, at least a portion of the boundary **112** of the mold cavity **114** may sealingly contact the surface **202** of the component **200**, such that the surface **202** of the component **200** also forms a portion of the boundary **112** of the mold cavity **114**.

[0028] In some embodiments described herein, the first and second brackets **104** and **106**, respectively, may be substantially rectangular in shape such that the body **102** has an overall body length $L_{sub.102}$ extending between a front end **120** and an opposite back end **122**, an overall body width $W_{sub.102}$ extending between a first side **124** and an opposite second side **126**, and an overall height $H_{sub.102}$ extending between outer surfaces **128** of each of the first and second brackets **104** and **106**, respectively. The first bracket **104** and the second bracket **106** may have a thickness $t_{sub.102}$. In some embodiments, the thickness $t_{sub.102}$ tapers, e.g., decreases, from the back end **122** towards the front end **120**. In some embodiments, a thickness $t_{sub.104}$ of the first

bracket **104** is different than a thickness $t_{sub.106}$ of the second bracket **106**. The body **102** may be variably sized, e.g., the body length $L_{sub.102}$ and the body width $W_{sub.102}$, to facilitate transporting the mold assembly **100**, e.g., in a tool case or tool bag, to and from a worksite where the component **200** is located. For example, the body length $L_{sub.102}$ and the body width $W_{sub.102}$ each may be less than about 5 inches, less than about 10 inches, and/or between about 5 inches to about 10 inches.

[0029] In the exemplary embodiment, the body **102** also includes a middle bracket **130** that extends between the first and second brackets **104** and **106**, respectively, and that is between the front and back ends **120** and **122**, respectively, of the mold assembly **100**. In some embodiments, the middle bracket **130** is approximately half-way between the front and back ends **120** and **122**, respectively. The middle bracket **130** may also define at least a portion of the boundary **112** of the mold cavity **114**. For example, the second bracket **106** includes a second inner surface **132**, the first bracket **104** includes a first inner surface **134**, and the middle bracket **130** includes a middle inner surface **136**, and at least a portion of the first, second, and middle inner surfaces **132**, **134**, and **136**, respectively, may define a portion of the boundary **112** of the mold cavity **114**. In some embodiments, at least a portion of the first inner surface **134**, the second inner surface **132**, and/or the middle inner surface **136** may be substantially planar and/or include at least a portion that is substantially planar. In some embodiments, at least one of the first inner surface **134**, the second inner surface **132** and/or the middle inner surface **136** may be concave, and/or may include one or more sections that are arcuate that form at least a portion of the boundary **112**. For example, in the illustrated embodiment best seen in FIG. 1, the middle inner surface **136** is concave. In some alternative embodiments, the inner surfaces **132**, **134**, and/or **136**, may include a raised boundary (not shown) for use in engaging the surface of the component **200**.

[0030] The mold assembly **100**, in the exemplary embodiment, also includes a passage **140** defined within, and extending through, the body **102** from an inlet **142** formed on the outer surface **138** of the body **102** to an outlet **144** formed on the boundary **112**. See FIG. 8, for example. The solidifying compound **302** may flow through the body opening **108** and through the passage **140** to fill the mold cavity **114**. In some embodiments, a hose **150** may be coupled to the inlet **142** and/or to the passage **140** to facilitate injecting a solidifying compound **302**, in liquid form, through the hose **150** and through the passage **140** to fill the mold cavity **114**. In some embodiments, either the hose **150** and/or the body **102** includes a connection mechanism **152**. In some embodiments, the passage **140** may include threads and/or any other fastening mechanism used to connect the hose **150** to the passage **140**.

[0031] In some embodiments, the passage **140** extends through either the first bracket **104** and/or the second bracket **106**. In some alternative embodiments, the passage **140** may extend through the middle bracket **130**. In some embodiments, the hose **150** may be inserted into the inlet **142** of the passage **140**, to enable the solidifying compound **302** to be pushed through the hose **150** and passage **140**, into the mold cavity **114**. In some alternative embodiments, the passage **140** may include threads to enable the hose **150** to securely couple to the passage **140**. The passage **140** may have any suitable shape or size, which facilitates the passage **140** being connected to the hose **150**. For example, in some embodiments, the hose **150** and the passage **140** may be relatively sized to enable the hose **150** to frictionally engage with the passage **140** in a secure connection between the hose **150** and the passage **140**. In some embodiments, the hose **150** may be secured to the passage **140** using in-line or elbow-shaped push-fit connectors or compression fittings.

[0032] In some embodiments, the mold assembly **100** includes a seal **160** that at least partially defines the boundary **112** of the mold cavity **114**. In some embodiments, the seal **160** may be a continuous seal formed integrally with, and/or coupled to, the first, second, and middle inner surfaces **132**, **134**, and **136**, respectively, and extending therefrom around a perimeter of the first, second, and middle inner surfaces **132**, **134**, **136**. The seal **160**, in the exemplary embodiment, includes a first seal portion **162**, coupled to the first bracket **104**, a second seal portion **164**, coupled

to the second bracket **106**, and a middle seal portion **166** coupled to the middle bracket **130**. When the mold assembly **100** is coupled with the component **200**, the entire seal **160** is sealingly coupled against the surface **202** of a component **200**. The seal **160** includes an inner surface **168** that defines, at least partially, a portion of the boundary **112** of the mold cavity **114**. As such, a height $h_{sub.160}$ of the seal **160** forms, or defines, a thickness $t_{sub.300}$ of the surface replica **300**, as best seen in FIG. **11**. In some embodiments, inner surfaces **132**, **134**, and/or **136** may include a chamfered portion that tapers against the seal **160**.

[0033] In some alternative embodiments, the mold assembly **100** includes two separate seals **160**, for example, the mold assembly **100** may include an upper seal, not shown, formed on the first inner surface **134** and a separate lower seal, not shown formed on the second inner surface **132**. In this alternative embodiment, when the mold assembly **100** is sealingly coupled against the component **200**, a portion of the upper and lower seals are positioned in scaling contact against the component **200**, creating one or more separate surface replicas and/or a single surface replica covering corners, edges, and/or irregular edge-shaped components or features.

[0034] In some embodiments, the seal **160** may be fabricated from a flexible material, e.g., rubber, rubber compound, or an elastomeric material, which compresses when a pressure and/or a force is applied. The seal **160** may be fabricated from any other suitable material, e.g., cork, silicone, neoprene, Teflon, plastic, and/or any other materials that enables the seal **160** to function as described herein. In some embodiments, the seal **160** may be fabricated from a different material than the rest of the body **102**, e.g., the first, second, and/or middle brackets **104**, **106**, and **130**, respectively. When the mold assembly **100** is coupled and clamped against the component **200**, the seal **160** sealingly engages the surface **202** of the component **200**.

[0035] With reference to FIGS. **5** and **8**, in some embodiments, the adjustment mechanism **110** may include a threaded bolt **170** that extends between the first bracket **104** and the second bracket **106**. The adjustment mechanism **110** may include a threaded bore **172** defined within at least one of the first bracket **104** and/or the second bracket **106**, and a through bore formed in the other of the first bracket **104** or second bracket **106**. The threaded bolt **170** may be coupled within the threaded bore **172**. Tightening the bolt **170** causes the first bracket **104** and the second bracket **106** to move towards each other, thus narrowing the distance d defined between the first and second brackets **104** and **106**, respectively. Likewise, loosening of the bolt **170** may cause the first bracket **104** and the second bracket **106** to move away from each other, thus increasing the distance d defined between the first and second brackets **104** and **106**, respectively. In some embodiments, tightening the bolt **170** causes the first bracket **104** to be pulled towards the second bracket **106** inducing a compressive force on the middle bracket **130**. The middle bracket **130** may have a thickness $t_{sub.130}$ that is relatively thin, and/or the middle bracket **130** may have suitable material properties, that causes the middle bracket **130** to function as a biasing mechanism, and the compressive force causes the middle bracket **130** to bend and/or compress, to narrow the distance d defined between the first and second brackets **104** and **106**, respectively. Likewise, in such an embodiment, loosening of the bolt **170** decreases the compressive force induced on the middle bracket **130** and the middle bracket **130** returns to its original length, restoring the distance d defined between the first and second brackets **104**, **106**.

[0036] In some embodiments, the adjustment mechanism **110** includes a handle **180** that may be used to selectively actuate the adjustment mechanism **110**. In the illustrated embodiment, the handle **180** includes a knob **182** that a user may grip to selectively tighten or loosen the bolt **170**. In other embodiments, the handle **180** may have any other suitable shape and/or size that enables the adjustment mechanism **110** to function as described herein. The handle **180** may be used to selectively adjust the adjustment mechanism **110** such that the first bracket **104** and the second bracket **106** may be moved towards each other or away from each other without the use of an additional tool. For example, the adjustment mechanism **110** may be used to couple the mold assembly **100** to a component **200** without the use of any other tool.

[0037] In some alternative embodiments, the adjustment mechanism **110** may include any other suitable adjustment mechanism, such as spring-assisted clamps and press-fit clamps. For example, adjustment mechanism **110** may include a biasing mechanism, e.g., a torsion or helical spring (not shown), that biases the upper and second bracket **106** towards each other. In some alternative embodiments, the adjustment mechanism **110** may include a rail mechanism (not shown) enabling translation between the first bracket **104** and second bracket **106**. In some alternative embodiments, the adjustment mechanism **110** may include a hinge (not shown) between the first bracket **104** and the second bracket **106** that enables rotation between the first bracket **104** and the second bracket **106**.

[0038] In some alternative embodiments, the adjustment mechanism **110** includes a clamp (not shown) that is connected with a threaded portion of the bolt **170** such that rotations of the clamp cause translations of the clamp relative to the threaded bolt **170**. Tightening of the clamp causes the clamp to press against the first bracket **104**, causing the first bracket **104** to move closer to the second bracket **106** such that the distance *d* is narrowed.

[0039] In reference to FIGS. **9** and **10**, the mold assembly **100** may be used to create a surface replica **300** of a surface **202** of a component **200**, e.g., turbine blade **201**. FIG. **9** shows the mold assembly **100** coupled to the surface **202** at an edge **204** of a turbine blade **201**. FIG. **10** shows another exemplary component **200**, embodied as a turbine blade **201**, showing various locations along the edge **204** of the turbine blade **201** where the mold assembly **100** may be sealingly coupled to the component **200**, to create a surface replica **300**. For example, the mold assembly **100** may be sealingly coupled to a tip portion **206**, a mid-portion **208**, and/or a root portion **210** of a leading edge **212** of the turbine blade **201**, for example. In some other embodiments, the mold assembly **100** may be sealingly engaged with a trailing edge **214** of the turbine blade **201**. In alternative embodiments, the mold assembly **100** may be sealingly coupled to any suitable surface **202** of any suitable component **200** for forming a surface replica **300** of the surface **202**.

[0040] In reference to FIG. **11**, an exemplary surface replica **300** formed using the mold assembly **100** is shown. The solidifying compound **302**, initially in a liquid form, may be injected into the mold cavity **114** such that the compound **302** surrounds and contacts the surface **202** of the component **200**. The solidified compound **302** forms the surface replica **300** with a first side **304** formed, at least partially, by the first bracket **104** and/or the first seal portion **162** of the seal **160**. Similarly, a second side **306** of the surface replica **300** is formed at least partially by the second bracket **106** and/or the second seal portion **164** of the seal **160**. The surface replica **300** includes a middle portion **308** formed at least partially by the middle bracket **130** and the middle seal portion **166**. The first side **304** includes a first inner surface replica **312**, the second side **306** includes a second inner surface replica **314**, and the middle portion **308** includes a middle inner surface replica **316**. The first inner surface replica **312** replicates a first side of the component **200** and the second inner surface replica **314** replicates an opposing second side of the component **200**. The middle inner surface replica **316** replicates at least a portion of the edge **204** of the component **200**.

[0041] The surface replica **300** may be formed of a solidifying compound **302** shaped by the boundary **112** and the surface **202**. In the exemplary embodiment, the mold assembly **100**, and in particular, the boundary **112** of the mold cavity **114** form surface replicas **300** which have a suitable thickness $t_{sub.300}$, e.g., a substantially uniform thickness $t_{sub.300}$ on opposite sides of the surface replica **300**, and/or a thickness $t_{sub.300}$ suitable for enabling the surface replica **300** to capture surface features, such as, but not limited to pitting, erosion, cavitation, rubbing, impacts, nicks, dents, and any other indications, for inspection. In some embodiments, the solidifying compound **302** may be a fast-curing silicone rubber having two primary components: a polymer, and a curing agent. The components are mixed together immediately prior to applying the solidifying compound **302** to the surface **202** of the component **200** within the mold cavity **114**. A thickness $t_{sub.304}$ of the first side **304** and a thickness $t_{sub.306}$ of the second side **306** are generally constant. For example, $t_{sub.304}$ and $t_{sub.306}$ are generally constant along a length

L.sub.300 of the surface replica **300**. In some embodiments, the thickness t.sub.304 and t.sub.306 may be approximately the same. A thickness t.sub.308 of the middle portion **308** may also be approximately the same as the thickness t.sub.304 and t.sub.306.

[0042] FIG. **12** illustrates an exemplary injector tool **400** for use with the mold assembly **100**. The injector tool may include a reservoir **402** that stores the solidifying compound **302** in a liquid form. The injector device includes an actuator **404** and a trigger **406** that selectively cause the actuator **404** to force or inject, the solidifying compound **302** through the hose **150**, through the passage **140**, and into the mold cavity **114**. In other embodiments, the mold assembly **100** may be used with any suitable device for supplying the solidifying compound **302** through the passage **140** formed through the body **102**. In some embodiments, a mold system may include the mold assembly **100**, the injector tool **400**, the solidifying compound **302** and/or the hose **150**.

[0043] FIG. **13** is a process flow diagram of an example method **500** that may be used to create the surface replica **300** using the mold assembly **100** shown in FIG. **1**. In some embodiments, the method includes adjusting **502** the adjustment mechanism **110** to increase the distance between the first bracket **104** and the second bracket **106** such that the distance between the first bracket **104** and the second bracket **106** is wide enough such that at least a portion of a component **200** may be inserted through the body opening **108** to be disposed between the first bracket **104** and the second bracket **106**. For example, in some embodiments, the method includes rotating the threaded bolt **170** in a first direction thereby reducing the compressive force across the first bracket **104**, second bracket **106**, and middle bracket **130** causing the first and second brackets **104**, **106** to move away from each other.

[0044] The method includes inserting **504** at least a portion of the component **200** through the body opening **108**, such that the component **200** is disposed between the first bracket **104** and the second bracket **106**. In some embodiments, the method **500** may include inserting the component **200** within the mold cavity **114** until the component **200** comes into contact with the middle seal portion **166**, such that the component **200** sealingly engages with the middle seal portion **166**. Since the middle seal portion **166** extends away, e.g., inward, the component **200** may be spaced a separation distance between the middle bracket **130**, such that solidifying compound **302** may flow between the component **200**, e.g., an edge **204** of the component **200**, and the middle inner surface **136**.

[0045] The method **500** also includes adjusting **506** the adjustment mechanism **110** to decrease the distance between the first bracket **104** and the second bracket **106** and seal the mold assembly **100** to the component **200**. The method **500** includes adjusting the adjustment mechanism **110** until the seal **160** contacts, and/or becomes sealingly engaged with the surface **202** of the component **200**. For example, the adjustment mechanism **110** may be used to adjust the distance d between the first bracket **104** and the second bracket **106** until the first seal portion **162** and the second seal portion **164** presses against the surface **202** of the component **200** and the seal **160** compresses. For example, in some embodiments, the method **500** includes rotating the threaded bolt **170** in a second direction, opposite the first direction, increasing the compressive force across the first bracket **104**, second bracket **106**, and middle bracket **130** causing the first and second brackets **104**, **106** to move towards each other and pressing the seal **160** against the surface **202** of the component **200**.

[0046] The method **500** includes injecting **508** into the mold cavity **114** the solidifying compound **302**. In some embodiments, the method **500** includes connecting the hose **150** to the passage **140**. For example, the method **500** may include inserting at least a portion of the hose **150** into the passage **140**. The method may also include injecting the solidifying compound **302**, e.g., using the injection tool, in liquid form through the hose **150** and through the passage **140** to fill the mold cavity **114**.

[0047] After the solidifying compound **302** solidifies, the method **500** may also include adjusting **510** the adjustment mechanism **110** to release the mold from the component **200**. For example, the adjustment mechanism **110** is used to widen the distance d between the first bracket **104** and the

second bracket **106**, as described above. The method **500** may also include **512** removing the surface replica **300** from the mold cavity **114** and/or off the surface **202** of the component **200**. The method **500** may also include inspecting the surface replica **300**, e.g., using an optical microscope, or any other suitable optical device.

[0048] Embodiments of the mold assembly described herein enable the creation of surface replicas that are consistent with a substantially constant thickness and with a suitable thickness that enables the surface replicas to capture surface defects. In some embodiments, the mold assembly includes an adjustment mechanism that enables the mold assembly to be selectively sealingly coupled or uncoupled from different or various locations on a single component or different sized components. In some embodiments, the mold may be selectively coupled to or uncoupled from the component without the use of any tools, such as wrenches or drills, by adjusting the adjustment mechanism.

[0049] Further aspects of the present disclosure are provided by the subject matter of the following clauses: [0050] 1. A mold for use in creating a surface replication of a surface of a component, the mold comprising: a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface, wherein the first and second inner surfaces define at least a portion of a boundary of a mold cavity; and an adjustment mechanism for selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component. [0051] 2. The mold in accordance with any preceding clause, wherein the body includes a passage defined within at least one of the first bracket and the second bracket such that the passage extends between an inlet and an outlet defined through the boundary of the mold cavity. [0052] 3. The mold in accordance with any preceding clause, wherein the body further comprises a middle bracket extending between the first bracket and the opposing second bracket, the middle bracket includes a middle inner surface that defines a portion of the boundary of the mold cavity. [0053] 4. The mold in accordance with any preceding clause, wherein the adjustment mechanism includes a threaded bore extending at least partially through at least one of the first bracket and the second bracket, and a threaded bolt sized to couple within the threaded bore such that rotation of the threaded bolt relative to the threaded bore selectively adjusts the distance defined between the first and second brackets. [0054] 5. The mold in accordance with any preceding clause, where in the body further includes a seal extending from the inner surfaces of the first bracket and the second bracket, the seal defines at least a portion of the boundary of the mold cavity. [0055] 6. The mold in accordance with any preceding clause, wherein the body further includes a seal extending from the first bracket, the second bracket, and the middle bracket, the seal defines at least partially a portion of the boundary of the mold cavity. [0056] 7. The mold in accordance with any preceding clause, wherein at least one of the first inner surface, the second inner surface, and the middle inner surface defining a portion of the boundary is concave. [0057] 8. The mold in accordance with any preceding clause, wherein the adjustment mechanism comprises a hinge rotationally coupling the first bracket and the second bracket and a biasing mechanism for selectively adjusting rotations of the hinge to enable the distance between defined between the first bracket and the second bracket to be selectively adjusted. [0058] 9. The mold in accordance with any preceding clause, wherein the first and second brackets are composed of a first material and the seal is composed of a second material that is different than the first material. [0059] 10. A mold system for use in creating a surface replication of a surface of a component, the mold system comprising: a reservoir for use in storing a solidifying compound; an injection tool fluidically connected to the reservoir; and a mold comprising: a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface, wherein the first and second inner surfaces define at least a portion of a boundary of a mold cavity; and an adjustment mechanism for

selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component. [0060] 11. The mold system in accordance with any preceding clause, wherein the body includes a passage defined within at least one of the first bracket and the second bracket such that the passage extends between an inlet and an outlet defined through the boundary of the mold cavity. [0061] 12. The mold system in accordance with any preceding clause, wherein the body further comprises a middle bracket extending between the first bracket and the opposing second bracket, the middle bracket includes a middle inner surface that defines a portion of the boundary of the mold cavity. [0062] 13. The mold system in accordance with any preceding clause, wherein the adjustment mechanism includes a threaded bore extending at least partially through at least one of the first bracket and the second bracket, and a threaded bolt sized to couple within the threaded bore such that rotation of the threaded bolt relative to the threaded bore selectively adjusts the distance defined between the first and second brackets. [0063] 14. The mold system in accordance with any preceding clause, wherein the body further includes a seal extending from the inner surfaces of the first bracket and the second bracket, the seal defines at least a portion of the boundary of the mold cavity. [0064] 15. The mold system in accordance with any preceding clause, wherein the body further includes a seal extending from the first bracket, the second bracket, and the middle bracket, the seal defines at least partially a portion of the boundary of the mold cavity. [0065] 16. The mold system in accordance with any preceding clause, wherein at least one of the first inner surface, the second inner surface, and the middle inner surface defining a portion of the boundary is concave. [0066] 17. The mold system in accordance with any preceding clause, wherein the adjustment mechanism comprises a hinge rotationally coupling the first bracket and the second bracket and a biasing mechanism for selectively adjusting rotations of the hinge to enable the distance between defined between the first bracket and the second bracket to be selectively adjusted. [0067] 18. The mold system in accordance with any preceding clause, wherein the first and second brackets are composed of a first material and the seal is composed of a second material that is different than the first material. [0068] 19. A method of creating a surface replication of a surface of a component, the method comprising: providing a mold assembly defining a mold cavity, the mold assembly including a first bracket, a second bracket, and an adjustment mechanism for selectively increasing or decreasing a distance defined between the first and second brackets; adjusting the adjustment mechanism to selectively increase a distance defined between the first bracket and the second bracket of the mold assembly; inserting a portion of a component between the first bracket and the second bracket; and adjusting the adjustment mechanism to cause the mold assembly to be in sealing contact with a portion of the surface of the component. [0069] 20. The method in accordance with any preceding clause, wherein the method comprises: injecting a solidifying compound into the mold cavity.

[0070] The above description is meant to be exemplary only, and one skilled in the art will recognize that changes may be made to the embodiments described without departing from the scope of the invention disclosed. Modifications, which fall within the scope of the present invention, will be apparent to those skilled in the art, in light of a review of this disclosure, and such modifications are intended to fall within the appended claims.

[0071] Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

[0072] This written description uses examples to disclose the embodiments of systems and methods, including the best mode, and also to enable any person skilled in the art to practice the systems and methods, including making and using any devices or systems and performing any

incorporated methods. The patentable scope of the systems and methods is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A mold for use in creating a surface replication of a surface of a component, the mold comprising: a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface, wherein the first and second inner surfaces define at least a portion of a boundary of a mold cavity; and an adjustment mechanism for selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component.
2. The mold in accordance with claim 1, wherein the body includes a passage defined within at least one of the first bracket and the second bracket such that the passage extends between an inlet and an outlet defined through the boundary of the mold cavity.
3. The mold in accordance with claim 1, wherein the body further comprises a middle bracket extending between the first bracket and the opposing second bracket, the middle bracket includes a middle inner surface that defines a portion of the boundary of the mold cavity.
4. The mold in accordance with claim 1, wherein the adjustment mechanism includes a threaded bore extending at least partially through at least one of the first bracket and the second bracket, and a threaded bolt sized to couple within the threaded bore such that rotation of the threaded bolt relative to the threaded bore selectively adjusts the distance defined between the first and second brackets.
5. The mold in accordance with claim 1, where in the body further includes a seal extending from the inner surfaces of the first bracket and the second bracket, the seal defines at least a portion of the boundary of the mold cavity.
6. The mold in accordance with claim 3, wherein the body further includes a seal extending from the first bracket, the second bracket, and the middle bracket, the seal defines at least partially a portion of the boundary of the mold cavity.
7. The mold in accordance with claim 3, wherein at least one of the first inner surface, the second inner surface, and the middle inner surface defining a portion of the boundary is concave.
8. The mold in accordance with claim 1, wherein the adjustment mechanism comprises a hinge rotationally coupling the first bracket and the second bracket and a biasing mechanism for selectively adjusting rotations of the hinge to enable the distance between defined between the first bracket and the second bracket to be selectively adjusted.
9. The mold in accordance with claim 5, wherein the first and second brackets are composed of a first material and the seal is composed of a second material that is different than the first material.
10. A mold system for use in creating a surface replication of a surface of a component, the mold system comprising: a reservoir for use in storing a solidifying compound; an injection tool fluidically connected to the reservoir; and a mold comprising: a body including a first bracket and an opposite second bracket spaced a distance from the first bracket, the first bracket includes a first inner surface and the second bracket includes a second inner surface, wherein the first and second inner surfaces define at least a portion of a boundary of a mold cavity; and an adjustment mechanism for selectively increasing or decreasing the distance defined between the first and second brackets, wherein the adjustment mechanism may selectively couple the first and second

brackets to the surface of the component in sealing contact such that the mold cavity is at least partially defined by the boundary of the mold and the surface of the component.

11. The mold system in accordance with claim 10, wherein the body includes a passage defined within at least one of the first bracket and the second bracket such that the passage extends between an inlet and an outlet defined through the boundary of the mold cavity.

12. The mold system in accordance with claim 10, wherein the body further comprises a middle bracket extending between the first bracket and the opposing second bracket, the middle bracket includes a middle inner surface that defines a portion of the boundary of the mold cavity.

13. The mold system in accordance with claim 10, wherein the adjustment mechanism includes a threaded bore extending at least partially through at least one of the first bracket and the second bracket, and a threaded bolt sized to couple within the threaded bore such that rotation of the threaded bolt relative to the threaded bore selectively adjusts the distance defined between the first and second brackets.

14. The mold system in accordance with claim 10, where in the body further includes a seal extending from the inner surfaces of the first bracket and the second bracket, the seal defines at least a portion of the boundary of the mold cavity.

15. The mold system in accordance with claim 12, wherein the body further includes a seal extending from the first bracket, the second bracket, and the middle bracket, the seal defines at least partially a portion of the boundary of the mold cavity.

16. The mold system in accordance with claim 12, wherein at least one of the first inner surface, the second inner surface, and the middle inner surface defining a portion of the boundary is concave.

17. The mold system in accordance with claim 10, wherein the adjustment mechanism comprises a hinge rotationally coupling the first bracket and the second bracket and a biasing mechanism for selectively adjusting rotations of the hinge to enable the distance between defined between the first bracket and the second bracket to be selectively adjusted.

18. The mold system in accordance with claim 14, wherein the first and second brackets are composed of a first material and the seal is composed of a second material that is different than the first material.

19. A method of creating a surface replication of a surface of a component, the method comprising: providing a mold assembly defining a mold cavity, the mold assembly including a first bracket, a second bracket, and an adjustment mechanism for selectively increasing or decreasing a distance defined between the first and second brackets; adjusting the adjustment mechanism to selectively increase a distance defined between the first bracket and the second bracket of the mold assembly; inserting a portion of a component between the first bracket and the second bracket; and adjusting the adjustment mechanism to cause the mold assembly to be in sealing contact with a portion of the surface of the component.

20. The method in accordance with claim 19, wherein the method comprises: injecting a solidifying compound into the mold cavity.
